📘 IB DIPLOMA PROGRAMME · 2024–2026 SPECIFICATION

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IB Math AA Complete Cheat Sheet

Analysis & Approaches · Full Syllabus · HL + SL

✦ All Key Formulas ✦ Worked Examples ✦ Exam Strategies ✦ HL Extensions

🔢 Algebra

📈 Functions

📐 Geometry & Trig

📊 Statistics

∫ Calculus

📖 GUIDE

How to Use This Cheat Sheet

Your 20-page system for fast revision and exam performance

🎯 PURPOSE

- → Fast-access formula reference during revision
- → Step-by-step method reminders
- → Exam tip and common-mistake alerts
- → Night-before final checklist

🗂️ ICON KEY

- 📐 **Formula** — core formula block
- ⚠️ **Common Mistake** — avoid this
- 💡 **Exam Insight** — what IB asks
- ⚡ **Quick Method** — shortcut
- 🟣 **HL Only** — higher level content
- ✅ **Tip** — best practice

📋 HOW TO REVISE

- → Start with the **Section Overview** pages
- → Test yourself on formulas before reading
- → Use **Worked Examples** to check method
- → Highlight any Exam Insights you've missed
- → Use **Page 20** the night before the exam

✅ **Best Practice:** Don't just read — cover the formula and try to recall it. Active recall is 3× more effective than passive reading.

🗺️ SYLLABUS MAP

All 5 topics of IB Math AA — pages and content overview

1 **Algebra — Pages 3–5**
Laws of exponents · Logarithms · Sequences & Series · Binomial theorem · Complex numbers (HL) · Proof (HL)

2 **Functions — Pages 6–7**
Function types · Transformations · Domain & range · Inverse functions · Rational functions · Composite functions

3 **Geometry & Trigonometry — Pages 8–9**
Trig ratios · Identities · Unit circle · Radians · Sine & cosine rule · 3D geometry · Vectors (HL)

4 **Statistics & Probability — Pages 10–11**
Descriptive stats · Probability rules · Binomial & normal distributions · Hypothesis testing (HL)

5 **Calculus — Pages 12–14**
Differentiation · Integration · Applications · Differential equations (HL) · Maclaurin series (HL)

HL **HL Extensions — Pages 15–16**
Complex numbers · Proof · Advanced calculus · Maclaurin series · Differential equations

★ **Exam Mastery — Pages 17–20**
Common mistakes · Strategy guide · Formula summary · Night-before checklist

💡 **Exam Insight:** IB examiners repeatedly test the same core skills. 80% of marks come from 20% of the syllabus. Focus on calculus applications, trig equations, and probability distributions.

SECTION 1 — ALGEBRA

Core Formulas

Exponents · Logarithms · Sequences · Binomial Theorem

Laws of Exponents

Rule	Formula
Product	$a^m \cdot a^n = a^{m+n}$
Quotient	$a^m / a^n = a^{m-n}$
Power	$(a^m)^n = a^{mn}$
Negative	$a^{-n} = 1/a^n$
Fractional	$a^{(m/n)} = \sqrt[n]{(a^m)}$
Zero	$a^0 = 1 \ (a \neq 0)$

⚠ **Mistake:** $(a+b)^n \neq a^n + b^n$ — this is the most common error!

Logarithm Laws

Law	Formula
Product	$\log(AB) = \log A + \log B$
Quotient	$\log(A/B) = \log A - \log B$
Power	$\log(A^n) = n \cdot \log A$
Change of Base	$\log_b(a) = \ln a / \ln b$
Inverse	$\log_a(a^x) = x$
Natural log	$\ln(e^x) = x$

KEY IDENTITY

$$y = \log_a(x) \iff x = a^y$$

Always convert log to exponential form to solve equations

Sequences & Series

ARITHMETIC SEQUENCE

$$u_n = u_1 + (n-1)d$$

$$S_n = n/2 \cdot (2u_1 + (n-1)d) = n/2 \cdot (u_1+u_n)$$

d = common difference; u_n = nth term

GEOMETRIC SEQUENCE

$$u_n = u_1 \cdot r^{n-1}$$

$$S_n = u_1(r^n-1)/(r-1) \text{ for } r \neq 1$$

$$S_\infty = u_1/(1-r) \text{ for } |r| < 1$$

r = common ratio; |r| < 1 for convergence

⚡ **Quick Method:** If sum to infinity exists → |r| < 1. Check this first!

Binomial Theorem

BINOMIAL EXPANSION

$$(a+b)^n = \Sigma \ C(n,r) \cdot a^{n-r} \cdot b^r$$

$$\text{General term: } T_{r+1} = C(n,r) \cdot a^{n-r} \cdot b^r$$

$$C(n,r) = n! / (r!(n-r)!)$$

r starts at 0; C(n,r) = "n choose r" — found on GDC or Pascal's triangle

💡 **Exam Insight:** IB frequently asks for a **specific term** or the **coefficient of x^k**. Set r+1 = term number, solve for r, substitute.

Methods & Techniques

Solving equations · Inequalities · Quadratics · Complex numbers

🔧 Solving Quadratics

QUADRATIC FORMULA

$$x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$$

$$\text{Discriminant: } \Delta = b^2-4ac$$

$\Delta > 0 \rightarrow$ 2 real roots | $\Delta = 0 \rightarrow$ 1 repeated root | $\Delta < 0 \rightarrow$ no real roots

SUM OF ROOTS

$$\alpha + \beta = -b/a$$

PRODUCT OF ROOTS

$$\alpha \cdot \beta = c/a$$

⚠️ **Mistake:** When dividing by a variable, check it ≠ 0 first — you may lose solutions!

📏 Solving Inequalities

→ Linear: solve like equation, flip sign when ×/÷ by negative
→ Quadratic: find roots, test sign in each interval
→ Rational: never cross-multiply — bring to one side first
→ Modulus: $|f(x)| < k \rightarrow -k < f(x) < k$
→ Sign diagram: (+)(−)(+) pattern alternates between roots

⚡ **Quick Method:** For quadratic inequality $ax^2+bx+c > 0$ with $a > 0$: answer is OUTSIDE the roots ($x < \alpha$ or $x > \beta$).

📐 Completing the Square

METHOD

$$ax^2 + bx + c = a(x + b/2a)^2 + (c - b^2/4a)$$

Vertex form: $a(x-h)^2 + k$, vertex = (h, k)

📉 Exponential & Log Equations

Method: Solving $a^x = b$ 1

→ Take ln both sides: $x \cdot \ln a = \ln b$
→ Solve: $x = \ln b / \ln a$
→ Or use GDC intersection method

Method: Solving log equations 2

→ Isolate the log expression
→ Convert to exponential: $\log_a(x) = b \rightarrow x = a^b$
→ Check solution (argument must be > 0)

🔢 Complex Numbers HL

FORMS & KEY FACTS

$$\text{Rectangular: } z = a + bi, \text{ where } i^2 = -1$$

$$\text{Modulus: } |z| = \sqrt{(a^2 + b^2)}$$

$$\text{Argument: } \arg(z) = \arctan(b/a) \text{ [adjust quadrant]}$$

$$\text{Polar: } z = r(\cos\theta + i \sin\theta) = re^{i\theta}$$

$$\text{Conjugate: } z^* = a - bi; \; z \cdot z^* = |z|^2$$

DE MOIVRE'S

$$z^n = r^n(\cos n\theta + i \sin n\theta)$$

EULER'S FORMULA

$$e^{i\theta} = \cos\theta + i \sin\theta$$

💡 **Exam Insight:** HL Paper 1 often asks to find roots of polynomials with complex roots, or to express z in polar form. Always check both a and b signs for correct quadrant.

📎 Proof by Induction HL

→ **Step 1:** Prove true for n = 1 (base case)
→ **Step 2:** Assume true for n = k
→ **Step 3:** Prove true for n = k+1 using assumption
→ **Step 4:** Conclude by induction

Worked Examples

Step-by-step IB-style solutions

EXAMPLE 1 — GEOMETRIC SERIES (SL)

The first term is 4, common ratio 1/2. Find S_∞.

1. Check: $|r| = 1/2 < 1$ ✓ (converges)
2. Apply: $S_\infty = u_1/(1-r) = 4/(1-½)$
3. $S_\infty = 4/(½) = 8$

$S_\infty = 8$

EXAMPLE 2 — BINOMIAL TERM (SL)

Find the coefficient of x^3 in $(2x - 1)^5$.

1. General term: T_{r+1} = C(5,r)·(2x)^(5−r)·(−1)^r
2. Need x^3: $5-r = 3$, so $r = 2$
3. $T_3 = C(5,2)\cdot(2x)^3\cdot(-1)^2 = 10\cdot 8x^3\cdot 1$

Coefficient = 80

EXAMPLE 3 — LOG EQUATION (SL)

Solve: $\log_2(x-1) + \log_2(x+1) = 3$

1. Combine: $\log_2((x-1)(x+1)) = 3$
2. Convert: $(x-1)(x+1) = 2^3 = 8$
3. $x^2 - 1 = 8 \rightarrow x^2 = 9 \rightarrow x = \pm 3$
4. Check: $x = -3 \rightarrow \log_2(-4)$ undefined ✗

$x = 3$

EXAMPLE 4 — ARITHMETIC SERIES (SL)

The sum of first n terms is $3n^2 + 5n$. Find u_1 and d.

1. $u_1 = S_1 = 3(1)^2 + 5(1) = 8$
2. $S_2 = 3(4) + 5(2) = 22$, so $u_2 = S_2 - S_1 = 14$
3. $d = u_2 - u_1 = 14 - 8 = 6$

$u_1 = 8$, $d = 6$

EXAMPLE 5 — COMPLEX NUMBERS (HL)

Express $z = 1 - i$ in polar form. Find z^3.

1. $|z| = \sqrt{(1^2 + (-1)^2)} = \sqrt{2}$
2. $\arg(z) = \arctan(-1/1) = -\pi/4$ (4th quadrant)
3. $z = \sqrt{2} \cdot e^{(-i\pi/4)}$
4. $z^3 = (\sqrt{2})^3\cdot e^{(-3i\pi/4)} = 2\sqrt{2}\cdot(\cos(-3\pi/4) + i\cdot\sin(-3\pi/4))$
5. $z^3 = 2\sqrt{2}\cdot(-\sqrt{2}/2 - i\cdot\sqrt{2}/2) = -2 - 2i$

$z^3 = -2 - 2i$

Exam Insight: In log equations, **always check your solution**: substitute back and confirm all log arguments are positive. Losing marks here is very common.

Quick Method — Quadratic in disguise: If you see $e^{2x} - 3e^x + 2 = 0$, let $u = e^x$ to get $u^2 - 3u + 2 = 0$, solve, then convert back.

SECTION 2 — FUNCTIONS

Types & Transformations

Function families · Transformations · Composite & inverse functions

Key Function Types

Type	Form	Key Feature
Linear	ax + b	Gradient = a
Quadratic	ax² + bx + c	Parabola, vertex = (-b/2a, ...)
Exponential	a^x, a > 0	Always positive; asymptote y=0
Logarithmic	log_a(x)	Inverse of exponential
Rational	p(x)/q(x)	Vertical asymptote: q(x)=0
Sine/Cosine	a·sin(bx+c)+d	Periodic, amplitude = |a|
Absolute	|f(x)|	Reflects negative part up

Transformations of f(x)

Transformation	Effect
f(x) + k	Shift UP k units
f(x) − k	Shift DOWN k units
f(x − h)	Shift RIGHT h units
f(x + h)	Shift LEFT h units
−f(x)	Reflect in x-axis
f(−x)	Reflect in y-axis
a·f(x) (a > 1)	Vertical stretch by a
f(bx) (b > 1)	Horizontal compress by 1/b

⚠️ **Mistake:** Horizontal shifts are OPPOSITE to the sign. f(x−3) shifts RIGHT, not left!

Composite & Inverse Functions

COMPOSITE

$(f \circ g)(x) = f(g(x))$

Apply g first, then f. Domain of f∘g = domain of g where g(x) is in domain of f.

INVERSE FUNCTION

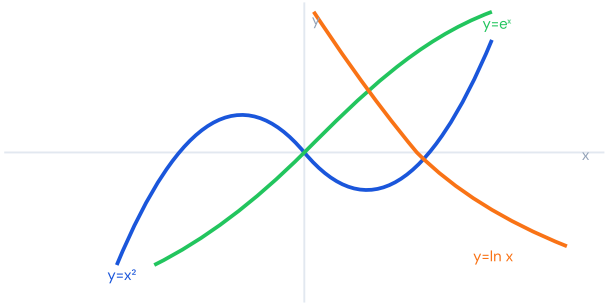
To find $f^{-1}(x)$:

1) Write y = f(x)
2) Swap x and y
3) Solve for y

Graph of f^{-1} is reflection of f in line y = x

✅ **Tip:** f and f^{-1} intersect on the line y = x. Domain of f^{-1} = Range of f.

Graphs of Common Functions



Asymptotes & Rational Functions

- → **Vertical asymptote:** x = a where denominator = 0
- → **Horizontal asymptote:** degree(num) < degree(den) → y = 0; equal degree → y = ratio of leading coefficients
- → **Oblique asymptote:** degree(num) = degree(den) + 1 → do polynomial division

💡 **Exam Insight:** For y = (ax+b)/(cx+d), horizontal asymptote is y = a/c. Vertical asymptote is x = −d/c.

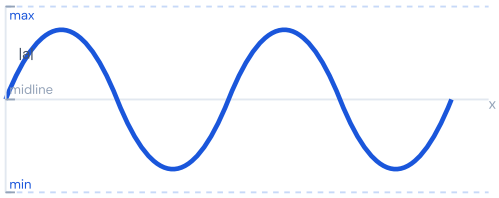
Graph Behavior & Domain/Range

Sine/cosine functions · Modulus · Domain & range rules

🌊 Sinusoidal Functions: y = a·sin(bx+c) + d

KEY PARAMETERS

Amplitude	= |a|
Period	= 2π/|b|
Phase shift	= -c/b (left if +, right if -)
Vertical shift	= d (midline)
Max value	= d + |a|
Min value	= d - |a|



📦 Domain & Range Quick Reference

Function	Domain	Range
√x	x ≥ 0	y ≥ 0
1/x	x ≠ 0	y ≠ 0
ln x	x > 0	ℝ
e^x	ℝ	y > 0
sin x	ℝ	-1 ≤ y ≤ 1
arcsin x	-1 ≤ x ≤ 1	-π/2 ≤ y ≤ π/2
arccos x	-1 ≤ x ≤ 1	0 ≤ y ≤ π
arctan x	ℝ	-π/2 < y < π/2

🧩 Modulus Function |f(x)|

→ |x| = x if x ≥ 0, |x| = -x if x < 0

→ Graph: reflect negative y-values in x-axis

→ |f(x)| = k → f(x) = k OR f(x) = -k

→ |f(x)| < k → -k < f(x) < k

→ |f(x)| > k → f(x) > k OR f(x) < -k

📊 Worked Example — Finding Range

📝 EXAMPLE — RANGE OF F(X) = 3SIN(2X) + 1

1. sin(2x) ranges from -1 to +1
2. 3·sin(2x) ranges from -3 to +3
3. 3·sin(2x) + 1 ranges from -2 to +4

Range: -2 ≤ f(x) ≤ 4

📝 EXAMPLE — TRANSFORMATIONS

f(x) = x². Describe g(x) = -f(x-2) + 3

1. f(x-2): shift right 2
2. -f(x-2): reflect in x-axis
3. +3: shift up 3

Vertex at (2, 3), opens downward

🔀 Even, Odd & Periodic Functions

EVEN: F(-X) = F(X)

Symmetric about y-axis. cos x, x^2, |x|

ODD: F(-X) = -F(X)

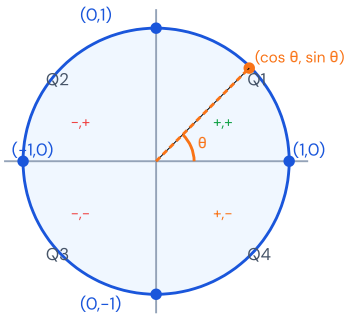
Symmetric about origin. sin x, x^3, tan x

💡 **Exam Insight:** Even/odd functions frequently appear in integration — ∫ odd dx from -a to a = 0. Use this to save time!

Trig Ratios & Identities

Unit circle · Identities · Sine & cosine rules · Area

⭕ Unit Circle



sin θ = y, cos θ = x, tan θ = y/x

📊 Exact Values Table

θ	0°	30°	45°	60°	90°
sin	0	½	√2/2	√3/2	1
cos	1	√3/2	√2/2	½	0
tan	0	√3/3	1	√3	–

⚡ **Quick Method:** The "finger rule" — for sin values, count fingers from left; square, divide by 4, take root. (0, ½, √2/2, √3/2, 1)

🔑 Key Identities

PYTHAGOREAN IDENTITIES

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta \quad \text{HL}$$

$$1 + \cot^2\theta = \csc^2\theta \quad \text{HL}$$

DOUBLE ANGLE FORMULAS

$$\sin(2\theta) = 2\cdot\sin\theta\cdot\cos\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$$

$$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta} \quad \text{HL}$$

COMPOUND ANGLE FORMULAS

$$\sin(A\pm B) = \sin A\cdot\cos B \pm \cos A\cdot\sin B$$

$$\cos(A\pm B) = \cos A\cdot\cos B \mp \sin A\cdot\sin B$$

$$\tan(A\pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\cdot\tan B}$$

🔪 Sine & Cosine Rules

SINE RULE

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Use when: given AAS, ASA, or SSA (ambiguous case)

COSINE RULE

$$a^2 = b^2 + c^2 - 2bc\cdot\cos A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Use when: given SAS or SSS

AREA OF TRIANGLE

$$\text{Area} = \tfrac{1}{2}\cdot a\cdot b\cdot\sin C$$

Also: Area = ½·base·height

⚠️ **Ambiguous Case (SSA):** Two triangles may exist! Always check if sin B > 1 (impossible), = 1 (one), or < 1 (check for second solution).

Radians, Equations & 3D Geometry

Radian measure · Solving trig equations · Vectors · 3D

Radian Measure

CONVERSION & ARC FORMULAS

$\text{rad} = \text{degrees} \times \pi/180$

$\text{degrees} = \text{rad} \times 180/\pi$

Arc length: $l = r\theta$ (θ in radians)

Sector area: $A = \frac{1}{2}r^2\theta$

Degrees	Radians	Exact
0°	0	0
30°	π/6	0.524
45°	π/4	0.785
60°	π/3	1.047
90°	π/2	1.571
180°	π	3.142
360°	2π	6.283

Solving Trig Equations

Step-by-step method

3

→ Isolate sin/cos/tan

→ Find principal value using inverse trig

→ Use symmetry to find all solutions in given interval

→ Add ±2πn or πn for general solution

SIN θ = K (GENERAL)

$\theta = \arcsin k + 2\pi n$ or $\pi - \arcsin k + 2\pi n$

COS θ = K (GENERAL)

$\theta = \pm\arccos k + 2\pi n$

⚡ **Quick Method:** CAST diagram — All (Q1), Sine (Q2), Tan (Q3), Cos (Q4). Use to find which quadrant gives a positive value.

3D Geometry

DISTANCE & MIDPOINT IN 3D

$|AB| = \sqrt{((x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2)}$

Midpoint $M = ((x_1+x_2)/2, (y_1+y_2)/2, (z_1+z_2)/2)$

→ Vectors SL HL

VECTOR OPERATIONS

$|v| = \sqrt{(v_1^2 + v_2^2 + v_3^2)}$

Unit vector: $\hat{v} = v/|v|$

Dot product: $a\cdot b = a_1b_1 + a_2b_2 + a_3b_3$

$a\cdot b = |a||b|\cos\theta \rightarrow \cos\theta = (a\cdot b)/(|a||b|)$

$a \perp b \iff a\cdot b = 0$

CROSS PRODUCT HL

$a\times b = |a||b|\sin\theta \cdot \hat{n}$

$|a\times b|$ = area of parallelogram

a×b perpendicular to both a and b

VECTOR EQUATION OF A LINE

$r = a + \lambda b$

a = position vector of point on line; b = direction vector; $\lambda \in R$

EXAMPLE — ANGLE BETWEEN VECTORS

Find angle between a=(1,2,−1) and b=(3,0,2)

1. $a\cdot b = (1)(3)+(2)(0)+(-1)(2) = 1$
2. $|a| = \sqrt{(1+4+1)} = \sqrt{6}$; $|b| = \sqrt{(9+0+4)} = \sqrt{13}$
3. $\cos\theta = 1/(\sqrt{6}\cdot\sqrt{13}) = 1/\sqrt{78}$

$\theta = \arccos(1/\sqrt{78}) \approx 83.5°$

💡 **Exam Insight:** For perpendicular/parallel lines with vectors, use dot product (perpendicular → 0, parallel → one is scalar multiple of other).

Descriptive Statistics & Probability

Mean · Variance · SD · Probability rules

Descriptive Statistics

MEASURES OF CENTRAL TENDENCY

Mean (discrete): $\bar{x} = \Sigma(f \cdot x) / \Sigma f$

Mean (continuous): $\bar{x} = \Sigma(f \cdot m) / \Sigma f$ (m = midpoint)

Median: middle value (50th percentile)

Mode: most frequent value

VARIANCE & STANDARD DEVIATION

Variance: $\sigma^2 = \Sigma f(x-\bar{x})^2 / \Sigma f$

Or: $\sigma^2 = (\Sigma f x^2)/n - \bar{x}^2$

SD: $\sigma = \sqrt{(\text{variance})}$

Use GDC for calculation in exam — know how to input data in 1-Var Stats

⚡ **Quick Method:** Always use GDC for σ, σ², $\bar{x}$. Enter in STAT → CALC → 1-Var Stats. Get all values instantly.

Correlation & Regression

PEARSON'S R & REGRESSION

$-1 \leq r \leq 1$ (correlation coefficient)

$r = 0$: no linear correlation

$r = \pm 1$: perfect linear correlation

Regression line: $\hat{y} = ax + b$ (passes through $(\bar{x}, \bar{y})$)

Use GDC: STAT → CALC → LinReg to find a, b, and r

⚠️ **Mistake:** Correlation ≠ causation. IB marks this distinction. Also: only predict within the data range (interpolation), not outside (extrapolation).

Probability Rules

CORE RULES

$0 \leq P(A) \leq 1$

$P(A') = 1 - P(A)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A|B) \cdot P(B)$

Conditional: $P(A|B) = P(A \cap B)/P(B)$

INDEPENDENCE

A,B independent $\iff P(A \cap B) = P(A) \cdot P(B)$

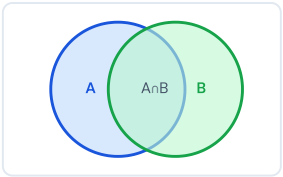
Also: $P(A|B) = P(A)$

Mutually exclusive: $P(A \cap B) = 0$

BAYES' THEOREM HL

$P(A|B) = P(B|A) \cdot P(A) / P(B)$

Venn Diagrams & Tree Diagrams



Tree diagram

EXAMPLE — CONDITIONAL PROBABILITY

P(A)=0.4, P(B)=0.5, P(A∩B)=0.2. Find P(A|B).

1. P(A|B) = P(A∩B)/P(B) = 0.2/0.5 = 0.4
2. Since P(A|B) = P(A) = 0.4, A and B are independent

P(A|B) = 0.4; A,B independent

Probability Distributions

Discrete distributions · Normal distribution · Hypothesis testing

Discrete Distributions

PROPERTIES OF ANY DISCRETE DISTRIBUTION

$\Sigma P(X = x) = 1$

$E(X) = \Sigma\, x \cdot P(X=x)$

$Var(X) = E(X^2) - [E(X)]^2$

$E(X^2) = \Sigma\, x^2 \cdot P(X=x)$

BINOMIAL DISTRIBUTION X~B(N,P)

$P(X=k) = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$

$E(X) = np$

$Var(X) = np(1-p)$

Conditions: fixed n, independent trials, constant p, 2 outcomes only

⚡ **Quick Method:** Use GDC: 2nd DISTR → binompdf(n,p,k) for P(X=k), binomcdf(n,p,k) for P(X≤k).

Poisson Distribution HL

X ~ PO(M)

$P(X=k) = e^{-m} \cdot m^k / k!$

$E(X) = Var(X) = m$

Conditions: events random, independent, uniform rate m = λt

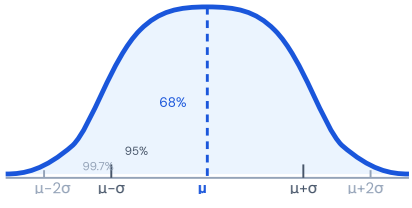
EXAMPLE — BINOMIAL

X~B(10, 0.3). Find P(X ≤ 3).

1. Use GDC: binomcdf(10, 0.3, 3)

P(X ≤ 3) ≈ 0.650

Normal Distribution X~N(μ, σ²)



KEY PROPERTIES

Mean = Median = Mode = μ

68% of data within μ ± σ

95% of data within μ ± 2σ

99.7% within μ ± 3σ

Standardize: Z = (X − μ)/σ, Z~N(0,1)

⚡ **Quick Method:** GDC: normalcdf(lower, upper, μ, σ) for probabilities. invNorm(p, μ, σ) for finding x given probability.

Hypothesis Testing HL

→ H_0: null hypothesis (no effect) vs H_1: alternative

→ p-value: probability of results given H_0 is true

→ Reject H_0 if p-value < significance level α

→ Type I error: reject H_0 when true (false positive)

→ Type II error: fail to reject H_0 when false (false negative)

💡 **Exam Insight:** IB commonly tests the normal distribution with "find the value of k given P(X < k) = 0.8". Always use invNorm on GDC.

∫ SECTION 5 — CALCULUS

Differentiation

Rules · Techniques · Chain/Product/Quotient rules

Core Differentiation Rules

f(x)	f'(x)
x^n	nx^{n-1}
e^x	e^x
e^{ax}	$a \cdot e^{ax}$
ln x	1/x
sin x	cos x
cos x	−sin x
tan x	$\sec^2 x = 1/\cos^2 x$
sin(ax)	a·cos(ax)
cos(ax)	−a·sin(ax)
a^x	$a^x \cdot \ln a$
arcsin x HL	$1/\sqrt{(1-x^2)}$
arccos x HL	$-1/\sqrt{(1-x^2)}$
arctan x HL	$1/(1+x^2)$

Advanced Rules

CHAIN RULE

$$d/dx[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Outside-inside: differentiate outer, keep inner, times derivative of inner

PRODUCT RULE

$$(uv)' = u'v + uv'$$

QUOTIENT RULE

$$(u/v)' = (u'v - uv') / v^2$$

Mnemonic: "lo d-hi minus hi d-lo, over lo squared"

IMPLICIT DIFFERENTIATION HL

Differentiate both sides w.r.t. x

d/dx[y] = dy/dx; d/dx[y²] = 2y·dy/dx

Collect dy/dx terms; solve

Applications of Differentiation

STATIONARY POINTS

→ Find f'(x) = 0 to locate stationary points

→ f''(x) > 0 → local minimum

→ f''(x) < 0 → local maximum

→ f''(x) = 0 → inconclusive → use sign diagram of f'

→ Inflection point: f''(x) = 0 AND changes sign

EQUATION OF TANGENT & NORMAL

Tangent at (a, f(a)): y − f(a) = f'(a)(x−a)

Normal: gradient = −1/f'(a)

Tangent touches curve; normal is perpendicular to tangent

RATES OF CHANGE

dy/dt = (dy/dx)·(dx/dt)

Chain rule for connected rates in context problems

EXAMPLE — CHAIN RULE

Find dy/dx for y = sin(3x²+1)

1. Outer: sin(u) → cos(u)
2. Inner: u = 3x²+1 → u' = 6x
3. dy/dx = cos(3x²+1)·6x

dy/dx = 6x·cos(3x²+1)

EXAMPLE — OPTIMISATION

Rectangle perimeter = 20cm. Maximise area.

1. 2l + 2w = 20 → l = 10 − w
2. A = l·w = (10−w)·w = 10w − w²
3. dA/dw = 10 − 2w = 0 → w = 5
4. d²A/dw² = −2 < 0 ✓ maximum

Max area = 25 cm² (square)

⚠ **Mistake:** Always verify maximum vs minimum using 2nd derivative test or sign chart. "Stationary" ≠ automatically "maximum"!

Integration

Standard integrals · Techniques · Definite integration

Standard Integrals

f(x)	∫f(x)dx
x^n $(n \neq -1)$	$x^{n+1}/(n+1) + C$
$1/x$	$\ln\|x\| + C$
e^x	$e^x + C$
e^{ax}	$e^{ax}/a + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sin(ax)$	$-\cos(ax)/a + C$
$\cos(ax)$	$\sin(ax)/a + C$
$1/(ax+b)$	$\ln\|ax+b\|/a + C$
$1/\sqrt{(1-x^2)}$ HL	$\arcsin x + C$
$1/(1+x^2)$ HL	$\arctan x + C$

⚠ **Mistake:** $\int 1/x^n\,dx \neq \ln(x^n)$. For $n \neq 1$: use power rule $x^{(1-n)}/(1-n)$. Only $\int 1/x = \ln|x|$.

Integration Techniques

INTEGRATION BY SUBSTITUTION

Let $u = g(x)$, $du = g'(x)dx$

$$\int f(g(x))\cdot g'(x)\,dx = \int f(u)\,du$$

Change all x's to u, including dx. Reverse substitute at end.

INTEGRATION BY PARTS HL

$$\int u\,dv = uv - \int v\,du$$

Choose u using LIATE: Logarithm, Inverse trig, Algebraic, Trig, Exponential

Definite Integration

FUNDAMENTAL THEOREM OF CALCULUS

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where F'(x) = f(x). Result is a number, not a function.

AREA BETWEEN CURVES

$$\text{Area} = \int_a^b |f(x) - g(x)|\,dx$$

$$\text{If } f(x) \geq g(x): \text{ Area} = \int_a^b (f(x)-g(x))\,dx$$

If curves cross: split at intersection, integrate each part separately

AREA ENCLOSED BY X-AXIS

$$\text{If } f(x) \geq 0: \text{ Area} = \int_a^b f(x)\,dx$$

$$\text{If } f(x) \leq 0: \text{ Area} = -\int_a^b f(x)\,dx$$

EXAMPLE — SUBSTITUTION

Evaluate $\int x\cdot e^{(x^2)}\,dx$

1. Let $u = x^2$, $du = 2x\,dx \rightarrow x\,dx = du/2$
2. $\int e^u \cdot du/2 = \frac{1}{2} e^u + C$
3. Substitute back: $= \frac{1}{2} e^{(x^2)} + C$

$\frac{1}{2} e^{(x^2)} + C$

EXAMPLE — AREA BETWEEN CURVES

Find area between $y=x^2$ and $y=x$ (from $x=0$ to $x=1$).

1. $x \geq x^2$ on $[0,1]$, so integrate $(x - x^2)$
2. $\int_0^1 (x - x^2)\,dx = [x^2/2 - x^3/3]_0^1$
3. $= (\frac{1}{2} - \frac{1}{3}) - 0 = 1/6$

Area = 1/6 sq units

⚡ **Quick Method:** Always sketch curves before integration to determine which is "on top". Intersection points are your limits.

💡 **Exam Insight:** IB often pairs integration with areas — worth 7–10 marks on Paper 2. Show all working including limits substituted in clearly.

∫ SECTION 5 — CALCULUS

Calculus Applications

Kinematics · Rates · Volumes · Differential equations

🚀 Kinematics

DISPLACEMENT, VELOCITY, ACCELERATION

$$v = \frac{ds}{dt} = s'(t)$$

$$a = \frac{dv}{dt} = v'(t) = s''(t)$$

$$s = \int v\,dt + C$$

$$v = \int a\,dt + C$$

Speed = |v|; Object at rest when v = 0; Direction changes when v changes sign

DISPLACEMENT VS DISTANCE

$$\text{Displacement} = \int_a^b v(t)\,dt \quad \text{(signed)}$$

$$\text{Distance} = \int_a^b |v(t)|\,dt \quad \text{(always positive)}$$

Split integral at t values where v = 0 to find distance

⚠️ **Mistake:** Distance ≠ |displacement|! If object reverses direction, integrate each segment separately and add absolute values.

📊 Graphs of f, f', f''

→ f'(x) = 0 → stationary point on f

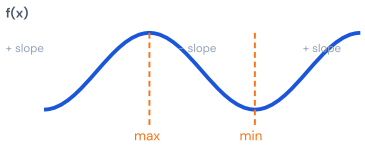
→ f'(x) > 0 → f is increasing

→ f'(x) < 0 → f is decreasing

→ f''(x) > 0 → f is concave UP (cup ∪)

→ f''(x) < 0 → f is concave DOWN (cap ∩)

→ f''(x) = 0 & sign change → inflection point on f



🔄 Differential Equations HL

SEPARABLE DES

$$\frac{dy}{dx} = f(x)\cdot g(y)$$

$$\text{Separate: } \int \frac{dy}{g(y)} = \int f(x)\,dx$$

Integrate both sides; add constant C

Apply initial conditions to find C

📝 EXAMPLE — SEPARABLE DE (HL)

Solve dy/dx = 2xy, y(0) = 3

1. Separate: dy/y = 2x dx
2. Integrate: ln|y| = x² + C
3. y = Ae^(x²), A = e^C
4. y(0)=3: 3 = A·e⁰ = A

y = 3e^(x²)

📐 Volume of Revolution HL

VOLUME AROUND X-AXIS

$$V = \pi \int_a^b [f(x)]^2\,dx$$

Rotate region under y=f(x) around the x-axis

⚡ Euler's Method (Numerical) HL

ITERATIVE FORMULA

$$y_{n+1} = y_n + h\cdot f(x_n, y_n)$$

$$x_{n+1} = x_n + h$$

h = step size (smaller h → more accurate)

💡 **Exam Insight:** HL Paper 3 often has a long question on DEs or Euler's method. Practice setting up the table step-by-step. Show each iteration clearly.

✅ **Tip:** For kinematics: ALWAYS state units with your answer. The question often specifies metres and seconds — use them consistently.

HL EXTENSIONS

Complex Numbers & Proof

Higher Level only — Pages 15–16

🔢 Complex Numbers — Complete Reference

ALL FOUR FORMS

Rectangular: $z = a + bi$

Polar: $z = r(\cos\theta + i\sin\theta) = r\cdot\text{cis}\,\theta$

Exponential: $z = re^{i\theta}$

where $r = |z| = \sqrt{a^2+b^2}$, $\theta = \arg(z)$

OPERATIONS IN POLAR FORM

$z_1z_2 = r_1r_2 \cdot \text{cis}(\theta_1+\theta_2)$

$z_1/z_2 = r_1/r_2 \cdot \text{cis}(\theta_1-\theta_2)$

$z^n = r^n \cdot \text{cis}(n\theta)$ [De Moivre]

NTH ROOTS OF Z

n roots: $r^k = r^{(1/n)} \cdot \text{cis}((\theta+2\pi k)/n)$

for $k = 0, 1, 2, \ldots, n-1$

Roots are equally spaced at $2\pi/n$ intervals on circle of radius $r^{(1/n)}$

EULER'S IMPORTANT IDENTITIES

$e^{i\pi} + 1 = 0$ (Euler's identity)

$\cos\theta = (e^{i\theta} + e^{-i\theta})/2$

$\sin\theta = (e^{i\theta} - e^{-i\theta})/2i$

📝 EXAMPLE — CUBE ROOTS OF 8

Find all cube roots of 8.

1. $8 = 8\cdot e^{i\cdot 0}$ in polar form
2. Roots: $8^{(1/3)}\cdot e^{i\cdot 2\pi k/3} = 2\cdot\text{cis}(2\pi k/3)$
3. k=0: 2, k=1: $2\text{cis}(2\pi/3) = -1+i\sqrt{3}$, k=2: $-1-i\sqrt{3}$

2, $-1+i\sqrt{3}$, $-1-i\sqrt{3}$

🔍 Types of Mathematical Proof

1. DIRECT PROOF

→ Start from assumption, derive conclusion logically

→ E.g. prove n^2+n always even: $n^2+n = n(n+1)$, product of consecutive integers

2. PROOF BY CONTRADICTION

→ Assume the negation is true

→ Derive a contradiction

→ Therefore original statement must be true

→ E.g. $\sqrt{2}$ is irrational — assume rational, derive contradiction

3. PROOF BY MATHEMATICAL INDUCTION

→ **Base case:** Verify for n = 1 (or n = 0)

→ **Inductive step:** Assume true for n = k

→ **Prove for n = k+1** using k assumption

→ **Conclusion:** "By the principle of mathematical induction, P(n) is true for all $n \in \mathbb{Z}^+$"

4. COUNTEREXAMPLE

→ Disprove a universal statement by finding ONE counterexample

→ E.g. "all primes are odd" → false (2 is prime and even)

💡 **HL Exam Insight:** Induction proofs must use the exact phrase "assume true for n = k" and "prove for n = k+1". The IB marking scheme awards specific marks for each step — don't skip any.

HL EXTENSIONS

Advanced Calculus

Maclaurin series · Advanced integration · L'Hôpital's rule

Maclaurin Series HL

GENERAL FORMULA

$$f(x) = \Sigma\, f^{(n)}(0)/n! \cdot x^n$$

$$= f(0) + f'(0)x + f''(0)x^2/2! + f'''(0)x^3/3! + \ldots$$

Function	Maclaurin Series	Valid for
e^x	$1 + x + x^2/2! + x^3/3! + \ldots$	all x
sin x	$x - x^3/3! + x^5/5! - \ldots$	all x
cos x	$1 - x^2/2! + x^4/4! - \ldots$	all x
ln(1+x)	$x - x^2/2 + x^3/3 - \ldots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + n(n-1)/2!\, x^2 + \ldots$	$\lvert x\rvert < 1$

⚡ **Quick Method:** For composite series (e.g. e^(x²)), substitute x^2 into known series for e^x — no need to differentiate repeatedly!

L'Hôpital's Rule HL

FOR 0/0 OR ∞/∞ FORMS

$$\lim f(x)/g(x) = \lim f'(x)/g'(x)$$

Only apply when limit gives 0/0 or ∞/∞ indeterminate form!

Advanced Integration Techniques HL

PARTIAL FRACTIONS

$p(x)/((x-a)(x-b)) = A/(x-a) + B/(x-b)$

Multiply through and equate coefficients or substitute x=a, x=b

Then integrate term by term

INTEGRATION BY PARTS (REPEATED)

$\int x^n e^x\, dx$: apply IBP n times

Tabular method: differentiate one, integrate other

IMPROPER INTEGRALS

$\int_1^{\infty} f(x)dx = \lim_{t\to\infty} \int_1^t f(x)dx$

Converges if limit exists; diverges if not

$\int_1^{\infty} 1/x^p\, dx$ converges iff $p > 1$

Second-Order DEs HL

HOMOGENEOUS: AY'' + BY' + CY = 0

Characteristic equation: $ar^2 + br + c = 0$

Two real roots: $y = Ae^{r_1 x} + Be^{r_2 x}$

Repeated root: $y = (A+Bx)e^{rx}$

Complex roots $\alpha \pm \beta i$: $y = e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

📝 **EXAMPLE — MACLAURIN SERIES**

Find first 3 non-zero terms of Maclaurin for f(x) = e^(sin x)

1. $\sin x \approx x - x^3/6 + \ldots$
2. Let $u = x - x^3/6$; $e^u \approx 1 + u + u^2/2 + \ldots$
3. $= 1 + (x - x^3/6) + (x - x^3/6)^2/2 + \ldots$
4. $= 1 + x + x^2/2 - x^3/6 + \ldots \approx 1 + x + x^2/2 + 0\cdot x^3$

$1 + x + x^2/2 - x^3/3 + \ldots$

💡 **HL Exam Insight:** Paper 3 is 1 hour, 55 marks, 2 long questions. It tests your ability to work through unfamiliar problems using core skills. Practice reading and understanding novel contexts — the math itself is always familiar.

⚠ EXAM MASTERY

15 Common Mistakes

Avoid these — they cost the most marks in IB exams

⚠ MISTAKE 1 — ALGEBRA

$(a+b)^2 = a^2 + 2ab + b^2$ is often written as $a^2 + b^2$

✓ Fix: Always expand fully. Middle term 2ab is critical.

⚠ MISTAKE 2 — LOGARITHMS

$\log(a+b) \neq \log a + \log b$

✓ Fix: Log product rule only works for multiplication: $\log(a \cdot b) = \log a + \log b$

⚠ MISTAKE 3 — DOMAIN OF LOG SOLUTIONS

Solving log equations and forgetting to check argument > 0

✓ Fix: Always verify: substitute back and check all log arguments are positive.

⚠ MISTAKE 4 — FUNCTION TRANSFORMATIONS

$f(x-3)$ shifts left (when it shifts RIGHT)

✓ Fix: Horizontal shifts are OPPOSITE to sign inside. $f(x-3) \rightarrow$ RIGHT 3.

⚠ MISTAKE 5 — TRIG EQUATIONS

Only finding one solution, missing solutions in other quadrants

✓ Fix: Always find all solutions in the given interval. Use CAST or unit circle.

⚠ MISTAKE 6 — CHAIN RULE

Differentiating $f(g(x))$ as $f'(g(x))$ without multiplying by $g'(x)$

✓ Fix: $d/dx[f(g(x))] = f'(g(x)) \times g'(x)$. Always multiply by inner derivative.

⚠ MISTAKE 7 — INTEGRATION CONSTANT

Forgetting +C in indefinite integration

✓ Fix: Always write +C. IB mark schemes deduct for missing constant.

⚠ MISTAKE 8 — DISTANCE VS DISPLACEMENT

Using $\int v\,dt$ to find distance when object changes direction

✓ Fix: Distance = $\int|v|\,dt$. Split integral at v=0 and add absolute values.

⚠ MISTAKE 9 — DISCRIMINANT

$b^2-4ac < 0$ means "no roots" — forgetting this means no REAL roots

✓ Fix: $\Delta < 0 \rightarrow$ no real roots (but still has complex roots in HL).

⚠ MISTAKE 10 — NORMAL DISTRIBUTION

Using wrong tail when $P(X > a)$ given instead of $P(X < a)$

✓ Fix: $P(X > a) = 1 - P(X < a)$. GDC normalcdf needs correct lower/upper bounds.

⚠ MISTAKE 11 — ROUNDING TOO EARLY

Rounding intermediate answers and losing accuracy in final answer

✓ Fix: Keep full precision on GDC throughout. Only round the FINAL answer to 3 s.f.

⚠ MISTAKE 12 — AMBIGUOUS CASE (SINE RULE)

Finding only one triangle when SSA gives two possibilities

✓ Fix: Check if the supplementary angle also gives a valid triangle (angle sum < 180°).

⚠ MISTAKE 13 — AREA BELOW X-AXIS

Integrating without absolute value when curve is below x-axis

✓ Fix: Area = $-\int f(x)dx$ when $f(x) \leq 0$. ALWAYS sketch first.

⚠ MISTAKE 14 — GEOMETRIC SERIES DIVERGENCE

Using S_∞ formula when $|r| \geq 1$ (series diverges)

✓ Fix: Always verify $|r| < 1$ before applying $S_\infty = u_1/(1-r)$.

⚠ MISTAKE 15 — INEQUALITY SIGN FLIP

Not reversing inequality when multiplying/dividing by a negative

✓ Fix: $-3x < 6 \rightarrow x > -2$ (sign FLIPS). Write a reminder to yourself in exam.

✅ EXAM MASTERY

Exam Strategy Guide

How to approach questions · Paper 1 vs Paper 2 tips

📋 Paper Structure Reminder

Paper	SL Time	HL Time	Calculator
Paper 1	90 min	120 min	❌ No
Paper 2	90 min	120 min	✅ Yes
Paper 3 (HL)	—	60 min	✅ Yes

✏️ Paper 1 — No Calculator

Non-Calculator Tactics

→ Memorise exact values (sin 30°, cos 45°, etc.)
→ Look for factoring, cancellation, clever substitution
→ Rationalise denominators; simplify before expanding
→ Avoid decimal answers — express exactly (√3/2, π, e)
→ Use CAST diagram for trig without GDC
→ Know log properties perfectly — they're tested here
→ Integration: check if you need a technique (sub, parts)

🖩 Paper 2 — Calculator

Calculator Tactics

→ Use GRAPH function to find intersections, zeros
→ STAT → CALC for regressions, 1-Var stats
→ DISTR for binomial and normal calculations
→ Store exact values in memory to avoid re-entry errors
→ For integrals: ∫f(x)dx on GDC can verify your answer
→ For maxima/minima: use CALC → max/min on graph
→ Never trust GDC blindly — always sanity check

🎯 Question Attack Strategy

Read the whole question first 1
Identify: topic, what is asked, what is given. Part (c) often relies on (a) and (b).

Write full working — even if guessing 2
IB awards method marks. A wrong answer with correct method = partial credit. A blank = zero.

Follow the mark count 3
[3 marks] means ~3 steps. If your solution has 1 line, you're missing something.

If stuck, move on and return 4
Don't spend 10 mins on 2 marks. Flag it, continue, return at end.

Check reasonableness 5
Does your answer make sense? Negative area? Impossible probability? Review.

🏆 High-Mark Section Strategy

✅ **Section B (long questions 7–10 marks):** These build progressively. Even if you can't complete the final part, the earlier parts earn most marks. Always attempt every part.

💡 **Exam Insight:** IB examiners say: "The most common reason for low marks is not attempting questions." A partially correct attempt always beats a blank. Write something mathematical for every part.

⚠️ **Significant Figures:** Always give answers to 3 significant figures unless told otherwise. "Exact" answers should stay exact (π, √2, fractions).

EXAM MASTERY

The 40 Most Important Formulas

Memorise these — they appear in nearly every IB Math AA exam

ALGEBRA

QUADRATIC FORMULA
$x=(-b\pm\sqrt{\Delta})/2a$

DISCRIMINANT
$\Delta = b^2-4ac$

GEOMETRIC NTH TERM
$u_n = u_1\cdot r^{n-1}$

GEOMETRIC SUM ∞
$S\infty = u_1/(1-r)$

BINOMIAL COEFF.
$C(n,r)=n!/(r!(n-r)!)$

BINOMIAL TERM
$T_{r+1}=C(n,r)a^{n-r}b^r$

CHANGE OF BASE
$\log_b a = \ln a/\ln b$

ARITHMETIC SUM
$S_n=n/2(2u_1+(n-1)d)$

GEOMETRY

SINE RULE
$a/\sin A=b/\sin B$

COSINE RULE
$a^2=b^2+c^2-2bc\cos A$

TRIANGLE AREA
$\frac{1}{2}ab\sin C$

ARC LENGTH
$l = r\theta$

SECTOR AREA
$A = \frac{1}{2}r^2\theta$

TRIGONOMETRY

PYTHAGOREAN ID
$\sin^2\theta+\cos^2\theta=1$

DOUBLE ANGLE SIN
$\sin 2\theta=2\sin\theta\cos\theta$

DOUBLE ANGLE COS
$\cos 2\theta=\cos^2\theta-\sin^2\theta$

COMPOUND SIN
$\sin(A+B)=\sin A\cos B+\cos A\sin B$

COMPOUND COS
$\cos(A+B)=\cos A\cos B-\sin A\sin B$

DOT PRODUCT
$a\cdot b=|a||b|\cos\theta$

STATISTICS

BINOMIAL E(X)
$E(X) = np$

BINOMIAL VAR(X)
$\mathrm{Var}(X) = np(1-p)$

BINOMIAL P(X=K)
$C(n,k)p^k(1-p)^{n-k}$

STANDARDISE
$Z=(X-\mu)/\sigma$

CONDITIONAL P
$P(A|B)=P(A\cap B)/P(B)$

ADDITION RULE
$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

VARIANCE
$\sigma^2=E(X^2)-[E(X)]^2$

∫ CALCULUS

POWER RULE (DIFF)
$d/dx[x^n]=nx^{n-1}$

CHAIN RULE
$f'(g(x))\cdot g'(x)$

PRODUCT RULE
$(uv)'=u'v+uv'$

QUOTIENT RULE
$(u/v)'=(u'v-uv')/v^2$

D/DX[Eˣ]
e^x

D/DX[LN X]
$1/x$

D/DX[SIN X]
$\cos x$

POWER RULE (INT)
$\int x^n=x^{n+1}/(n+1)+C$

∫EˣDX
e^x+C

∫1/X DX
$\ln|x|+C$

FTC
$\int_a^b f(x)dx=F(b)-F(a)$

INT. BY PARTS
$\int u\,dv=uv-\int v\,du$

VELOCITY
$v=ds/dt,\ a=dv/dt$

🌙 FINAL REVISION

Night Before Exam

Your 30-minute final checklist — stay calm, stay sharp

✅ Formula Recall Checklist

Cover each and try to recall before uncovering ✓

- [ ] Quadratic formula + discriminant rules
- [ ] Arithmetic & Geometric sequences (nth term + sum)
- [ ] Sum to infinity + convergence condition
- [ ] Binomial expansion + general term T_{r+1}
- [ ] All 6 log laws
- [ ] Exact trig values (0°,30°,45°,60°,90°)
- [ ] $\sin^2\theta + \cos^2\theta = 1$ and double angle formulas
- [ ] Sine rule, cosine rule, area = ½ab sinC
- [ ] Chain rule, product rule, quotient rule
- [ ] Derivatives: e^x, ln x, sin, cos, tan
- [ ] Standard integrals + substitution method
- [ ] Fundamental theorem of calculus
- [ ] Binomial distribution formula + E(X) + Var(X)
- [ ] Normal distribution Z-score standardisation
- [ ] Conditional probability + independence condition

> ✅ **If you can recall all 15 above without looking — you're ready for a 7.**

🧠 Key Concepts Summary — Topic by Topic

🔢 ALGEBRA

→ Logs and exponentials are inverses of each other
→ Binomial: find specific term by setting up T_{r+1}
→ Geometric series only converges if $|r| < 1$

📈 FUNCTIONS

→ Horizontal shifts are OPPOSITE to sign in bracket
→ Domain of f^{-1} = Range of f
→ Even functions: reflect in y-axis; Odd: rotate 180° about origin

📐 TRIG

→ Degrees to radians: ×π/180
→ Use CAST for trig signs in all quadrants
→ Check ambiguous case whenever SSA is given

📊 STATISTICS

→ Use GDC for all normal/binomial calculations
→ $P(X > a) = 1 - P(X \leq a)$
→ Correlation ≠ causation — state this explicitly

∫ CALCULUS

→ Always find +C in indefinite integration
→ Sketch curves before calculating areas
→ Max/min: $f'=0$ AND $f''\neq 0$ to confirm type

🌟 FINAL MINDSET

→ Show full working — every line can earn marks
→ Attempt every question — never leave a blank
→ Round only the final answer (3 s.f.)
→ Check with GDC where possible
→ You've got this. Good luck! 🎯